

Lab Manual – Classes

Student will learn how to build

- Pointer to Data member
- Pointer to member function
- Pointers object

Task 1

You are developing a program to manage a library's collection of books. Each book has a unique identifier, a title, an author, and a quantity available. You want to implement a system where the user can search for a book by its identifier and update the quantity available using a pointer to a data member.

Create a class Book that have following data members

- Int id
- String title
- String author_name
- Int quantity_available

Declare a pointer to a data member of the “Book” class that points to the “quantityAvailable” member.

Implement a function that searches for a book by its identifier and updates the quantity available using the pointer to the data member.

```
void UpdateBookQuantity(Book* books, int numBooks, int identifier, int newQuantity);
```

Task 2

You are developing a task management application that allows users to create and manage their tasks. Each task has a title, description, and a completion status. You want to implement a system where users can mark a task as complete or incomplete using a pointer to a member function

- Define a class called `Task` that represents a task and includes member variables for the task's title, description, and completion status.
- Create an instance of the `Task` class to represent a specific task in the task management application.

- Declare a pointer to a member function of the `Task` class that points to a member function responsible for updating the task's completion status.
- Implement two member functions in the `Task` class: `Complete()` and `Incomplete()`. These functions will update the `isComplete` member variable accordingly.
- Set the pointer to the appropriate member function based on the user's input.
- Call the member function pointed to by the `completionPtr` to update the task's completion status.

Task 3

You are developing a customer management system for a retail store. The system should allow the store manager to create and manage customer profiles. Each customer profile has a unique ID, name, email address, and phone number. You want to implement a system where the manager can dynamically create and store customer profiles.

Define a class called `Customer` that represents a customer profile. Include member variables for the customer's unique ID, name, email address, and phone number.

Implement a data structure, such as an array or a vector, to store the customer profiles.

Call the `CreateCustomerProfile()` function whenever the store manager wants to create a new customer profile, passing the necessary parameters such as ID, name, email, and phone number.